

```

from sklearn.base import BaseEstimator, TransformerMixin
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import MinMaxScaler, OrdinalEncoder
import joblib
import math

df = pd.read_csv("./diamonds.csv", index_col=0)

```

- price: price in US dollars (\$326-\$18,823) (numérique, target)
- carat: weight of the diamond (0.2-5.01) (numérique)
- cut: quality of the cut (Fair, Good, Very Good, Premium, Ideal) (ordinaire)
- color: diamond color (J (worst) to D (best)) (ordinaire)
- clarity: measurement of how clear the diamond is (I1 (worst), SI2, SI1, VS2, VS1, VVS2, VVS1, IF (best)) (ordinaire)
- x: length in mm (0-10.74) (numérique)
- y: width in mm (0-58.9) (numérique)
- z: depth in mm (0-31.8) (numérique)
- depth: total depth percentage (43-79) (numérique)
- table: width of top of diamond relative to widest point (43-95) (numérique)

```

target = "price"
num_columns = ["carat", "x", "y", "z", "depth", "table"]
ord_columns = ["cut", "color", "clarity"]

y = df[target]
X = df.drop(target, axis=1)

```

Après le premier train_test_split:

```

X      : #####
X_train : #####
X_test  :      ##

```

Après le second train_test_split:

```

X      : #####
X_train : #####
X_val   :      ##
X_test  :      ##

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size =
0.2, random_state=314)
#X_train, X_val, y_train, y_val = train_test_split(X_train, y_train,
test_size = 0.25)

```

```
print(X_train.shape)
#print(X_val.shape)
print(X_test.shape)
print(y_train.shape)
#print(y_val.shape)
print(y_test.shape)
```

```
(43152, 9)
(10788, 9)
(43152,)
(10788,)
```

Analyse des données

```
df.isna().sum()
```

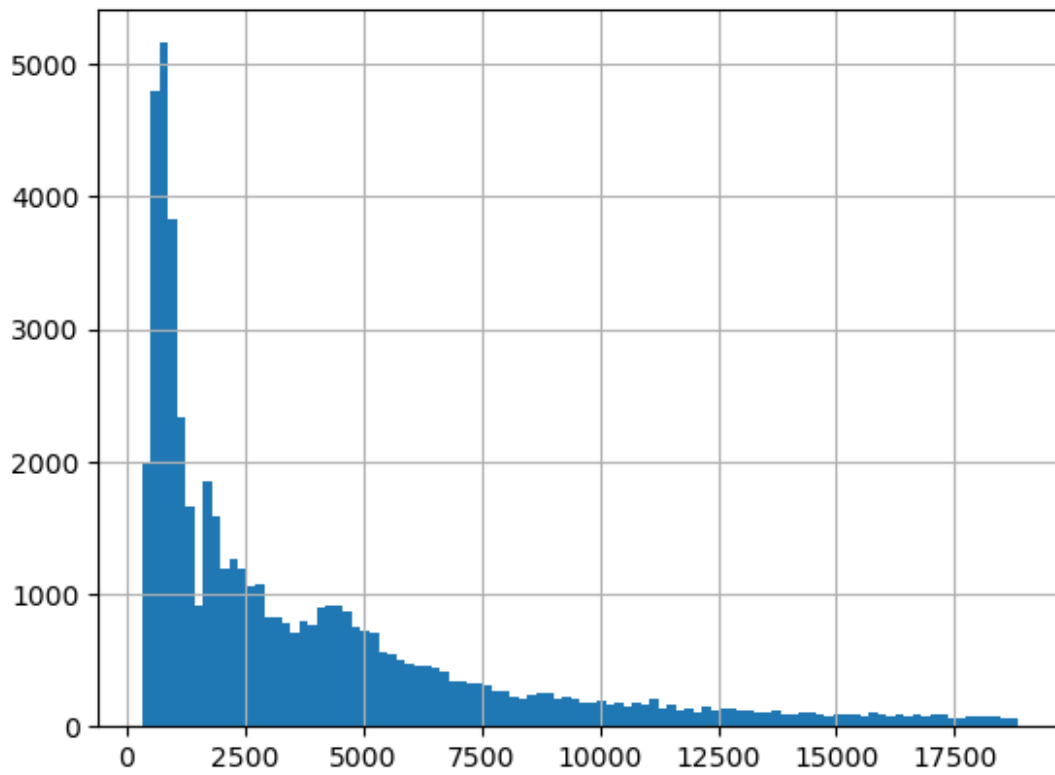
```
carat      0
cut         0
color       0
clarity     0
depth       0
table       0
price       0
x           0
y           0
z           0
dtype: int64
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 53940 entries, 1 to 53940
Data columns (total 10 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   carat       53940 non-null  float64
 1   cut         53940 non-null  object
 2   color       53940 non-null  object
 3   clarity     53940 non-null  object
 4   depth       53940 non-null  float64
 5   table       53940 non-null  float64
 6   price       53940 non-null  int64
 7   x           53940 non-null  float64
 8   y           53940 non-null  float64
 9   z           53940 non-null  float64
dtypes: float64(6), int64(1), object(3)
memory usage: 4.5+ MB
```

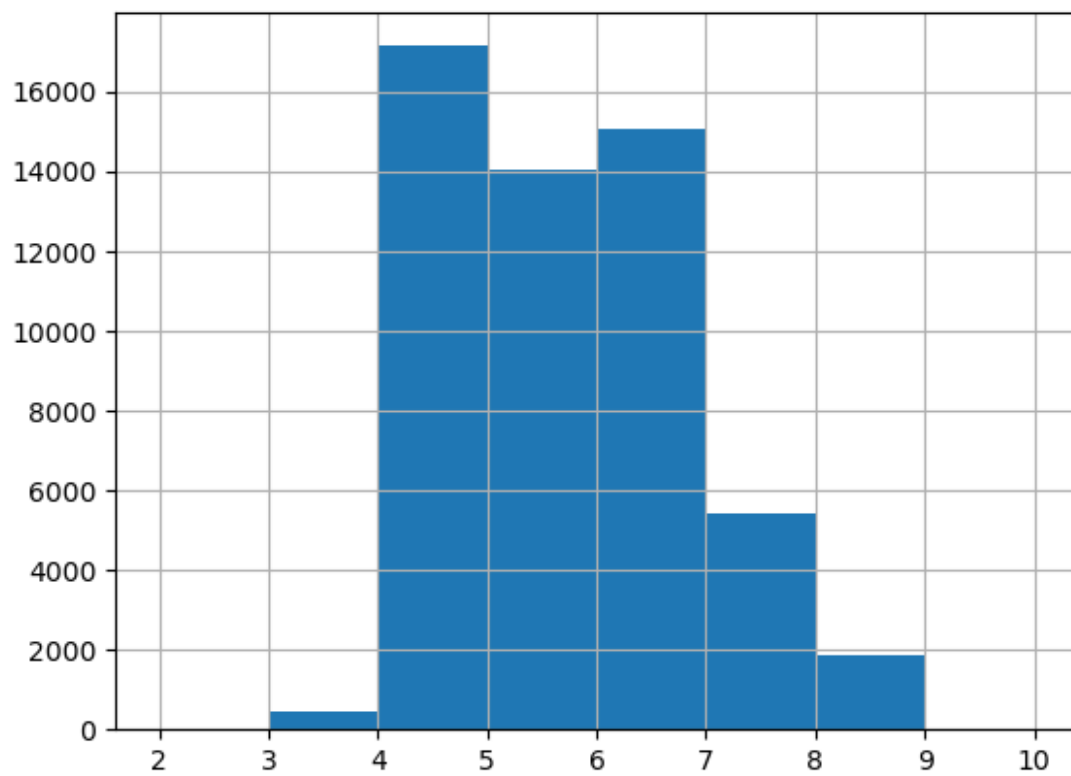
```
df["price"].hist(bins=100)
```

<AxesSubplot:>



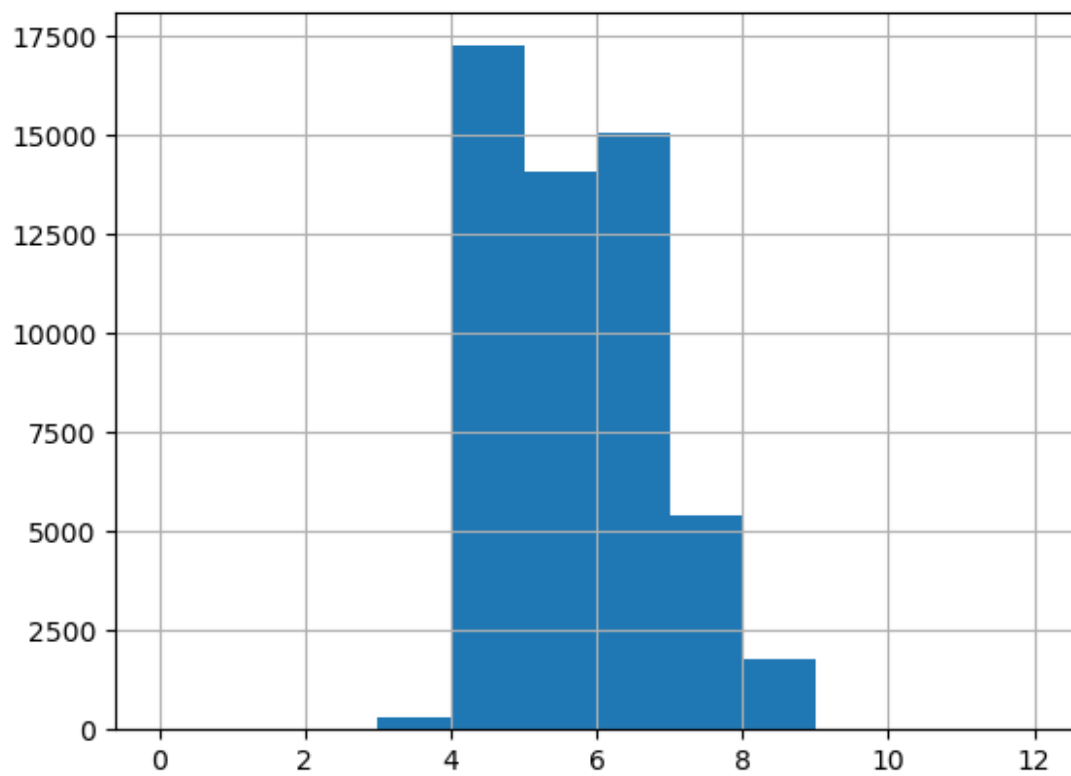
```
df["x"].hist(bins=range(2,11))
```

<AxesSubplot:>



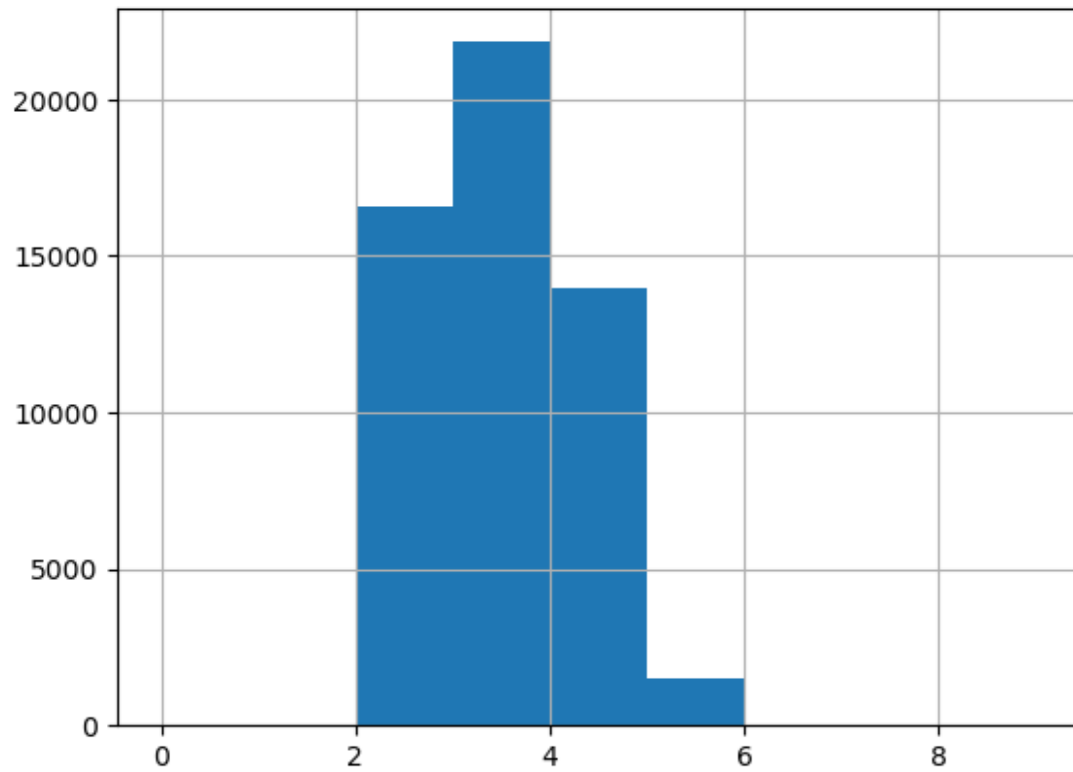
```
df["y"].hist(bins = range(0,13))
```

```
<AxesSubplot:>
```



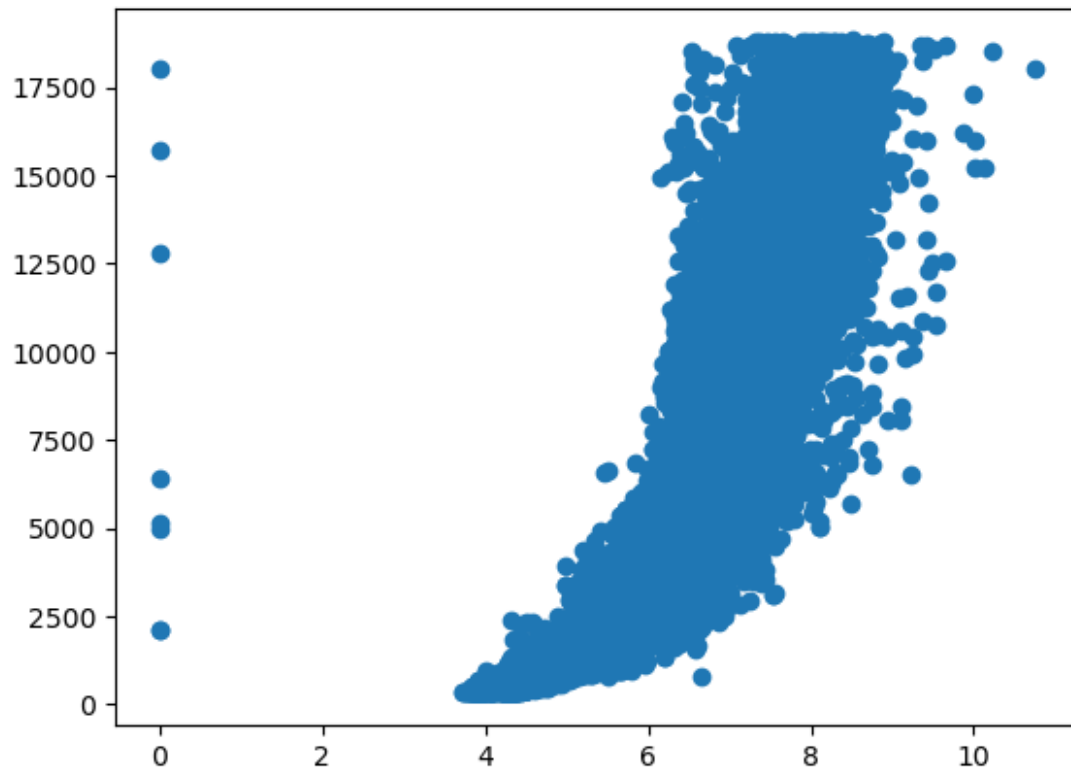
```
df["z"].hist(bins=range(0,10))
```

```
<AxesSubplot:>
```

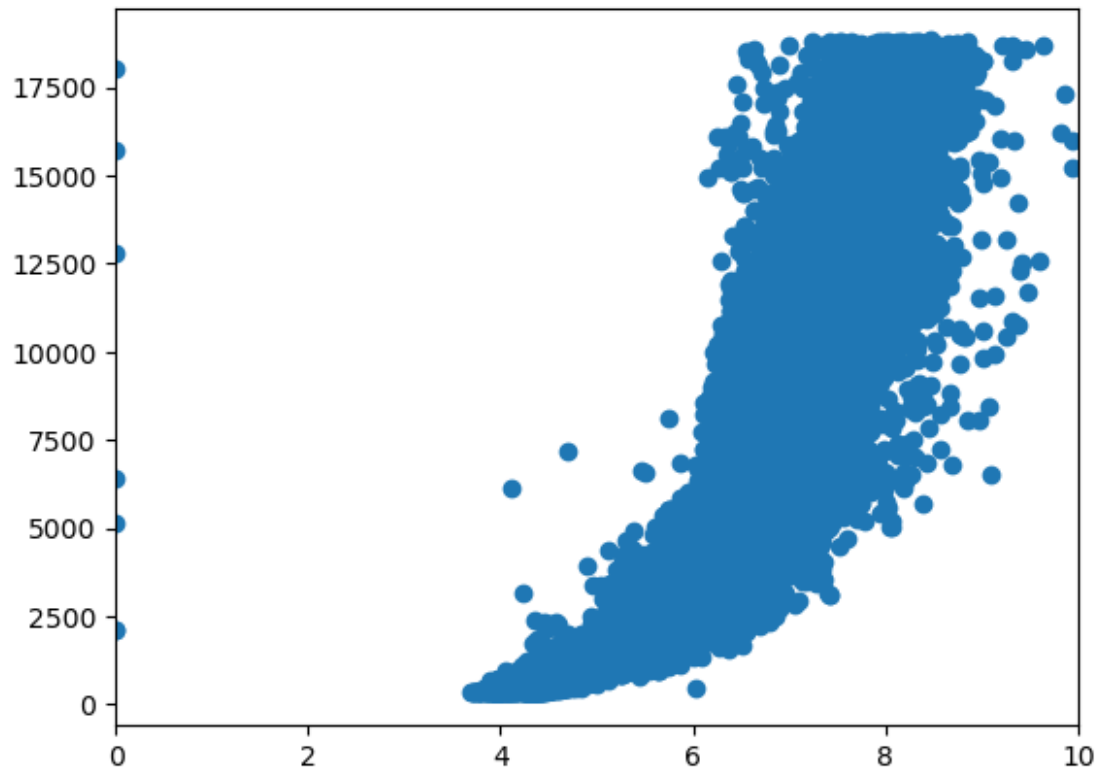


```
plt.scatter(df["x"], df["price"])
```

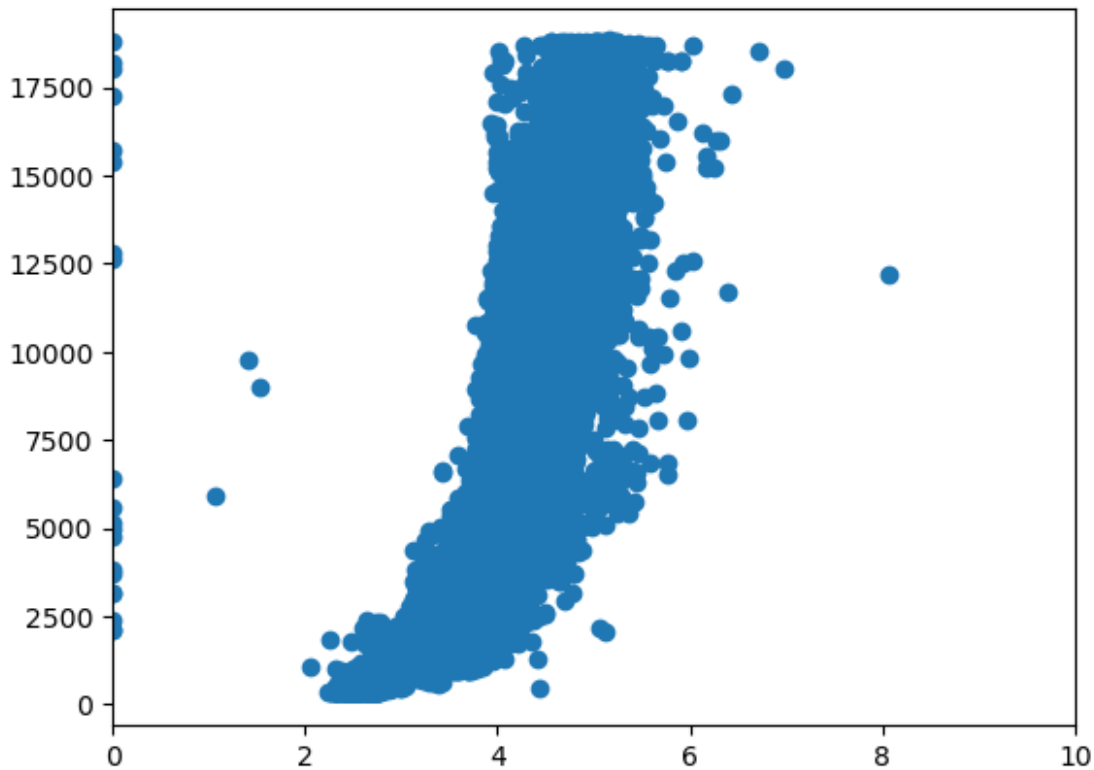
```
<matplotlib.collections.PathCollection at 0x7327cec082b0>
```



```
plt.scatter(df["y"], df["price"])  
plt.xlim((0, 10))  
(0.0, 10.0)
```



```
plt.scatter(df["z"], df["price"])  
plt.xlim((0, 10))  
(0.0, 10.0)
```

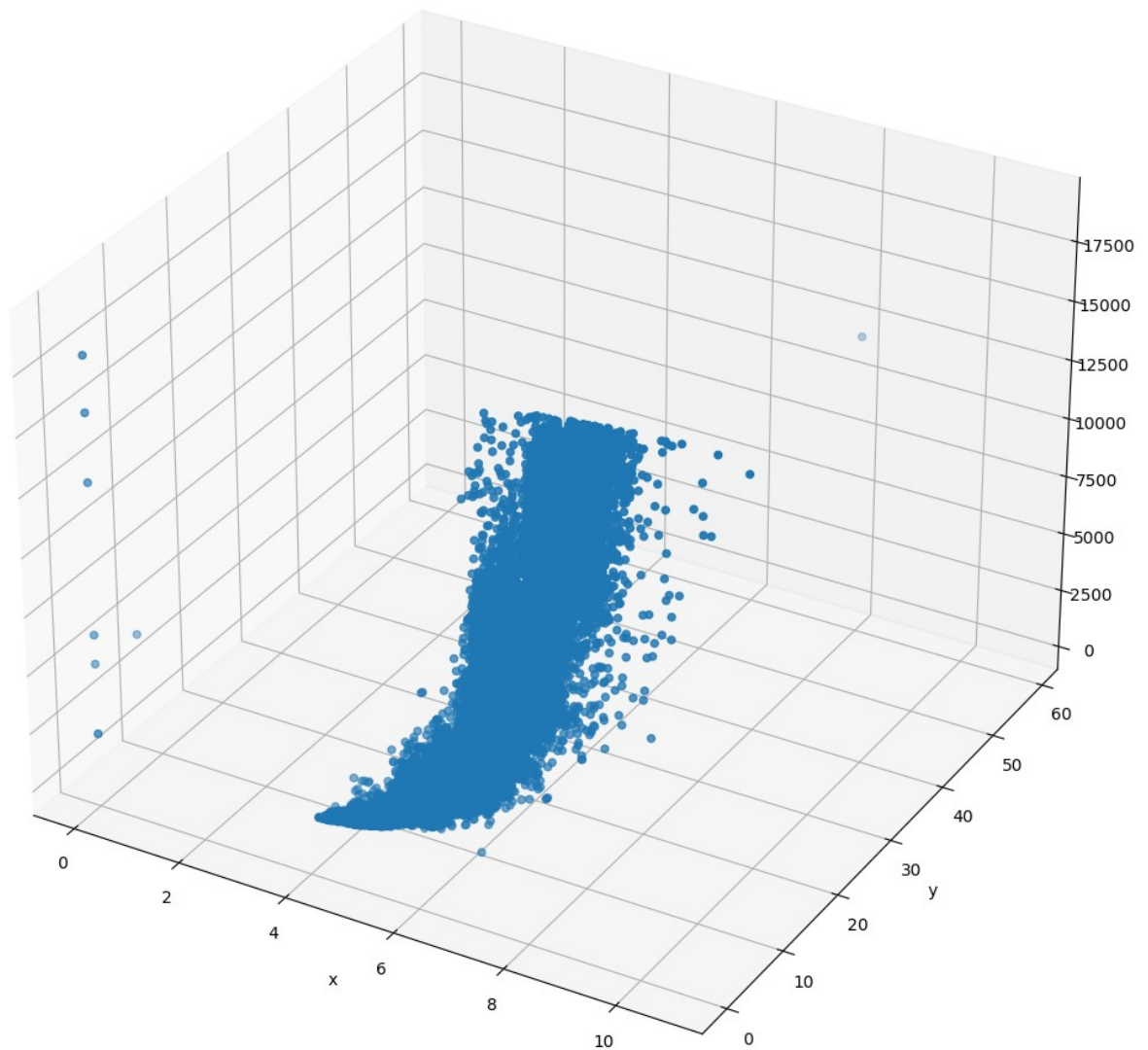



```
from mpl_toolkits.mplot3d import Axes3D

def plot_3D(df):
    fig = plt.figure(figsize = (10,10))
    ax = Axes3D(fig)
    ax.scatter(df["x"], df["y"], df["price"])
    plt.xlabel("x")
    plt.ylabel("y")
    plt.show()
```

```
plot_3D(df)
```

```
/tmp/ipykernel_359905/2826463635.py:3: MatplotlibDeprecationWarning:
Axes3D(fig) adding itself to the figure is deprecated since 3.4. Pass
the keyword argument auto_add_to_figure=False and use fig.add_axes(ax)
to suppress this warning. The default value of auto_add_to_figure will
change to False in mpl3.5 and True values will no longer work in 3.6.
This is consistent with other Axes classes.
  ax = Axes3D(fig)
```



```
df["cut"].value_counts()
```

Ideal	21551
Premium	13791
Very Good	12082
Good	4906
Fair	1610

Name: cut, dtype: int64

```
df["clarity"].value_counts()
```

SI1	13065
VS2	12258

```

SI2      9194
VS1      8171
VVS2     5066
VVS1     3655
IF        1790
I1        741
Name: clarity, dtype: int64

df["color"].value_counts()

G      11292
E      9797
F      9542
H      8304
D      6775
I      5422
J      2808
Name: color, dtype: int64

```

Prétraitement

Méthode pour détecter les anomalies numériques

- Calculer les quartiles : Q1 et Q3
- Calculer la distance interquartile : $\text{delta} = Q3 - Q1$
- Valeurs à garder : $[Q1 - 1.5 \times \text{delta} ; Q3 + 1.5 \times \text{delta}]$

```

class OutlierRemover(BaseEstimator, TransformerMixin):
    def __init__(self, columns, factor = 1.5, remove=True):
        self.columns = columns
        self.factor = factor
        self.remove = remove
        self.kept_indices = None

    def remove_outliers(self, X):
        l = []
        for c in self.columns:
            Q1 = X[c].quantile(0.25)
            Q3 = X[c].quantile(0.75)
            delta = Q3 - Q1
            lower_bound = Q1 - self.factor * delta
            upper_bound = Q3 + self.factor * delta
            column = X[c].loc[ ( X[c] >= lower_bound ) & ( X[c] <=
upper_bound ) ]
            l.append( pd.Series(column) )

        res = pd.concat(l, axis=1, join = "inner")

```

```

        self.kept_indices = res.index

        return X.loc[self.kept_indices]

    def fit(self, X, y=None):
        return self

    def transform(self, X):
        if self.remove:
            return self.remove_outliers(X)
        else:
            self.kept_indices = X.index
            return X

outlier_pipeline = Pipeline(steps = [("outliers",
OutlierRemover(columns=num_columns, factor=1.5, remove=False))])

color_pipeline = Pipeline(steps = [("encoder",
OrdinalEncoder(categories=[["J", "I", "H", "G", "F", "E", "D"]]))])
cut_pipeline = Pipeline(steps = [("encoder",
OrdinalEncoder(categories=[["Fair", "Good", "Very Good", "Premium",
"Ideal"]]))])
clarity_pipeline = Pipeline(steps = [("encoder",
OrdinalEncoder(categories=[["I1", "SI2", "SI1", "VS2", "VS1", "VVS2",
"VVS1", "IF"]]))])

num_preprocessor = ColumnTransformer([
    ("scaler", MinMaxScaler(), num_columns)
],
    remainder="passthrough",
    verbose_feature_names_out=False
)
num_preprocessor.set_output(transform="pandas")

ColumnTransformer(remainder='passthrough',
                    transformers=[('scaler', MinMaxScaler(),
                                  ['carat', 'x', 'y', 'z', 'depth',
'weight']),
                                ('depth', MinMaxScaler(),
                                  ['depth'])],
                    verbose_feature_names_out=False)

ord_preprocessor = ColumnTransformer([
    ("color", color_pipeline, ["color"]),
    ("cut", cut_pipeline, ["cut"]),
    ("clarity", clarity_pipeline, ["clarity"]),
],
    remainder="passthrough",
    verbose_feature_names_out=False
)
ord_preprocessor.set_output(transform="pandas")

```

```
ColumnTransformer(remainder='passthrough',  
                  transformers=[('color',  
                                Pipeline(steps= [('encoder',  
OrdinalEncoder(categories=[['J',  
  
    'I',  
  
    'H',  
  
    'G',  
  
    'F',  
  
    'E',  
  
    'D']])))),  
                               ('color',  
                                Pipeline(steps= [('encoder',  
OrdinalEncoder(categories=[['Fair',  
  
    'Good',  
  
    'Very Good',  
  
    'Good',  
  
    'Premium',  
  
    'Ideal'])])),  
                               ('cut',  
                                Pipeline(steps= [('encoder',  
OrdinalEncoder(categories=[['I1',  
  
    'SI2',  
  
    'SI1',  
  
    'VS2',  
  
    'VS1',  
  
    'VVS2',  
  
    'VVS1',
```

```
'IF']]]))],  
                                ['clarity']],  
                                verbose_feature_names_out=False)  
  
pipeline = Pipeline(steps=[  
    ("outliers", outlier_pipeline),  
    ("num", num_preprocessor),  
    ("ord", ord_preprocessor)  
],verbose = 4)  
  
pipeline  
Pipeline(steps=[('outliers',  
                 Pipeline(steps=[('outliers',  
                                 OutlierRemover(columns=['carat',  
'x', 'y',  
                                                    'z',  
'depth',  
                                                    'table'],  
                                                remove=False)))]),  
                ('num',  
                 ColumnTransformer(remainder='passthrough',  
                                   transformers=[('scaler',  
MinMaxScaler(),  
                                                  ['carat', 'x', 'y',  
'z',  
                                                  'depth',  
'table'])])),  
                verbose_feature_names_out=False)),  
            ('ord',  
             ColumnTransformer(remainder='passthrough',  
                               tran...  
OrdinalEncoder(categories=[[ 'J',  
'I',  
'H',  
'G',  
'F',  
'E',  
'D']])))],  
               ['color']),  
              ('cut',  
Pipeline(steps=[('encoder',
```

```

OrdinalEncoder(categories=[['Fair',
'Good',
'Very ',
'Good',
'Premium',
'Ideal']])))),
                                ['cut']),
                                ('clarity',

Pipeline(steps=[('encoder',
OrdinalEncoder(categories=[['I1',
'SI2',
'SI1',
'VS2',
'VS1',
'VVS2',
'VVS1',
'IF']])))),
                                ['clarity'])),
                                verbose_feature_names_out=False)),
                                verbose=4)

X_train_clean = pipeline.fit_transform(X_train)
y_train_clean =
y_train.loc[pipeline.named_steps["outliers"].named_steps["outliers"].k
ept_indices]

[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

```

```

X_test_clean = pipeline.transform(X_test)
y_test_clean =
y_test.loc[pipeline.named_steps["outliers"].named_steps["outliers"].ke
pt_indices]

print(X_train_clean.shape)
#print(X_val.shape)
print(X_test_clean.shape)
print(y_train_clean.shape)
#print(y_val.shape)
print(y_test_clean.shape)

(43152, 9)
(10788, 9)
(43152,)
(10788,)

```

Sauvegarde

```

X_train_clean.to_csv("./X_train.csv", index=False)
X_test_clean.to_csv("./X_test.csv", index=False)
y_train_clean.to_csv("./y_train.csv", index=False)
y_test_clean.to_csv("./y_test.csv", index=False)

joblib.dump(pipeline, "pipeline_preprocessing_diamants.joblib")

['pipeline_preprocessing_diamants.joblib']

```

Régression linéaire

```

from sklearn.metrics import mean_squared_error, r2_score,
mean_absolute_error
from sklearn.linear_model import LinearRegression

pipeline_final = Pipeline(steps = [
    ("preprocessing", pipeline),
    ("linear_regression", LinearRegression())
])

pipeline_final.fit(X_train, y_train_clean)

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(

```



```

[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.1s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

Pipeline(steps=[('preprocessing',
                  Pipeline(steps=[('outliers',
                                   Pipeline(steps=[('outliers',
OutlierRemover(columns=['carat',
                    'x',
                    'y',
                    'z',
                    'depth',
                    'table'])]))),
                  ('num',
ColumnTransformer(remainder='passthrough',
transformers=[('scaler',
MinMaxScaler(),
                ['carat',
                'x', 'y',
                'z',
                'depth',
                'table'])],
verbose_feature_names_out=False)),
                  ('ord',
ColumnTransformer(remainde...
                ['color'])),
                ('cut',
Pipeline(steps=[('encoder',
OrdinalEncoder(categories=[['Fair',
                            'Good',

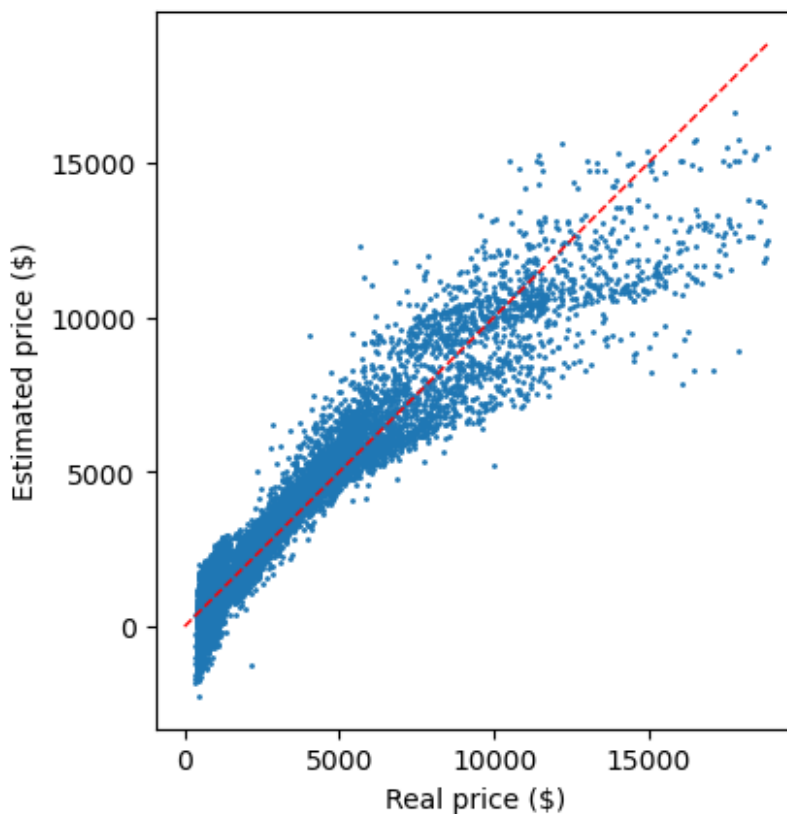
```



```
Mean Squared Error      : 1082591.288513738
Root Mean Squared Error : 1040.476471869373
Mean Absolute Error      : 705.5085261728541
R2                        : 0.9073453118386716
```

```
def plot_pred_vs_test(y_pred, y_test):
    plt.scatter(y_test, y_pred, s=1)
    maxi = max(y_test.max(), y_pred.max())
    #plt.xlim(0, maxi)
    #plt.ylim(0, maxi)
    ax = plt.gca()
    ax.set_aspect('equal', adjustable='box')
    plt.xlabel("Real price ($)")
    plt.ylabel("Estimated price ($)")
    plt.plot([0,maxi], [0, maxi], color = "r", ls="--", linewidth=1)
    plt.show()

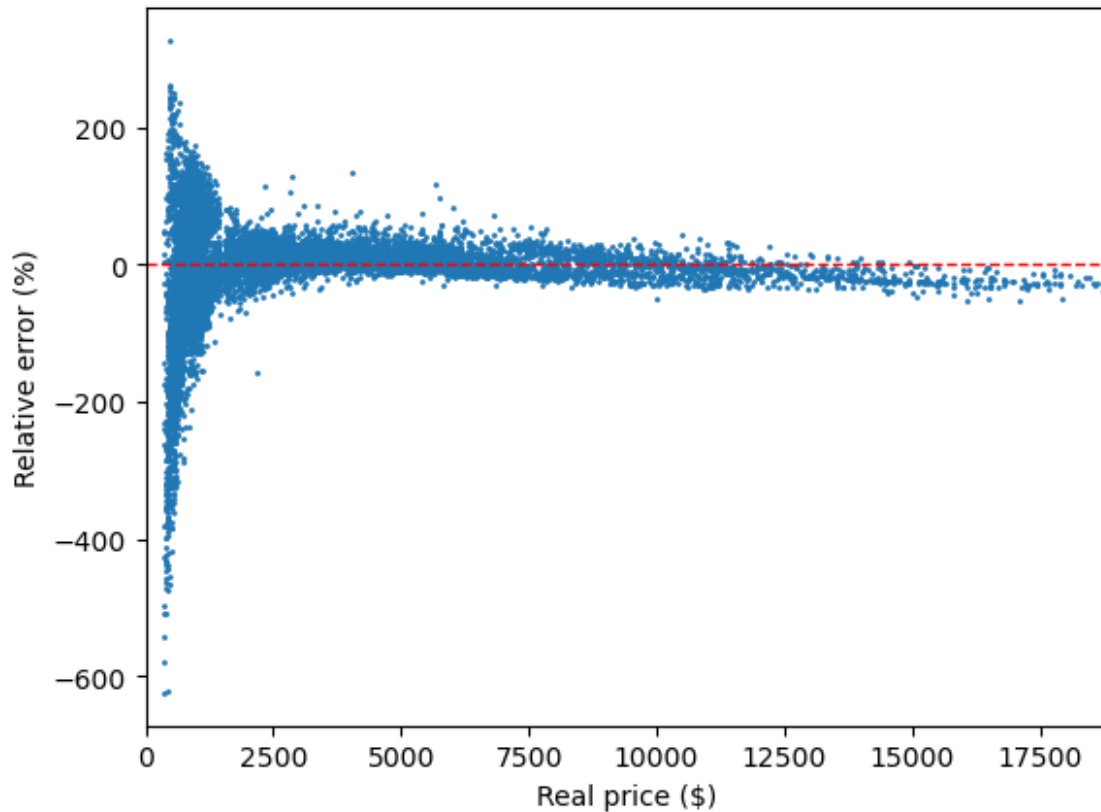
plot_pred_vs_test(y_pred, y_test_clean)
```



```
def plot_pred_vs_test_relative(y_pred, y_test):
    plt.scatter(y_test, 100*(y_pred-y_test)/y_test, s=1)
    maxi = y_test.max()
    plt.xlim(0, maxi)
    plt.xlabel("Real price ($)")
```

```
plt.ylabel("Relative error (%)")
plt.plot([0, maxi], [0, 0], color = "r", ls="--", linewidth=1)
plt.show()
```

```
plot_pred_vs_test_relative(y_pred, y_test_clean)
```



Estimation de la valeur d'un diamant

```
carat = 10
cut = "Ideal"
color = "E"
clarity = "VVS1"
x = 20
y = 20
z = 30
depth = 50
table = 90

mon_beau_diamant = pd.DataFrame([{
    "carat" : 10,
    "cut" : "Ideal",
    "color" : "E",
    "clarity" : "VVS1",
```

```

        "x" : 20,
        "y" : 20,
        "z" : 30,
        "depth" : 50,
        "table" : 90,
    })
mon_beau_diamant

```

	carat	cut	color	clarity	x	y	z	depth	table
0	10	Ideal	E	VVS1	20	20	30	50	90

```

pipeline_final.predict(mon_beau_diamant)[0]
7337.968839335017

```

Régression avec les k plus proches voisins

```

from sklearn.neighbors import KNeighborsRegressor
from sklearn.model_selection import GridSearchCV

pipeline_final_knn = Pipeline(steps = [
    ("preprocessing", pipeline),
    ("knn", KNeighborsRegressor())
])

hyperparameters_knn = {
    "preprocessing__outliers__outliers__remove": [False],
    "knn__n_neighbors" : range(2,10),
    "knn__weights" : ["uniform", "distance"],
}

```

GridSearchCV vise à maximiser la fonction de score. Dans un problème de régression, on souhaite minimiser l'erreur (MSE ou MAE). Il faut donc renseigner un scoring égal à l'opposé de l'erreur: "neg_mean_squared_error" ou "neg_mean_absolute_error"

```

gs_knn = GridSearchCV(
    estimator=pipeline_final_knn,
    param_grid=hyperparameters_knn,
    cv = 4,
    scoring = "neg_mean_squared_error",
    verbose = 10
)

gs_knn.fit(X_train, y_train)

```

Fitting 4 folds for each of 16 candidates, totalling 64 fits
 [CV 1/4; 1/16] START knn__n_neighbors=2, knn__weights=uniform,
 preprocessing__outliers__outliers__remove=False

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean parameter is deprecated in version 1.2 and won't be supported anymore in version 1.4.
```

```
warnings.warn(
```

```
[CV 1/4; 1/16] END knn__n_neighbors=2, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;, score=-664781.633
total time= 0.6s
```

```
[CV 2/4; 1/16] START knn__n_neighbors=2, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean parameter is deprecated in version 1.2 and won't be supported anymore in version 1.4.
```

```
warnings.warn(
```

```
[CV 2/4; 1/16] END knn__n_neighbors=2, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;, score=-719515.912
total time= 0.4s
```

```
[CV 3/4; 1/16] START knn__n_neighbors=2, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean parameter is deprecated in version 1.2 and won't be supported anymore in version 1.4.
```

```
warnings.warn(
```

```
[CV 3/4; 1/16] END knn__n_neighbors=2, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;, score=-637997.909
total time= 0.4s
```

```
[CV 4/4; 1/16] START knn__n_neighbors=2, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean
```

parameter is deprecated in version 1.2 and won't be supported anymore in version 1.4.

```
warnings.warn(
```

```
[CV 4/4; 1/16] END knn__n_neighbors=2, knn__weights=uniform,  
preprocessing__outliers__outliers__remove=False;; score=-712693.131  
total time= 0.4s
```

```
[CV 1/4; 2/16] START knn__n_neighbors=2, knn__weights=distance,  
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/  
_param_validation.py:591: FutureWarning: Passing an int for a boolean  
parameter is deprecated in version 1.2 and won't be supported anymore  
in version 1.4.
```

```
warnings.warn(
```

```
[CV 1/4; 2/16] END knn__n_neighbors=2, knn__weights=distance,  
preprocessing__outliers__outliers__remove=False;; score=-633534.541  
total time= 0.5s
```

```
[CV 2/4; 2/16] START knn__n_neighbors=2, knn__weights=distance,  
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/  
_param_validation.py:591: FutureWarning: Passing an int for a boolean  
parameter is deprecated in version 1.2 and won't be supported anymore  
in version 1.4.
```

```
warnings.warn(
```

```
[CV 2/4; 2/16] END knn__n_neighbors=2, knn__weights=distance,  
preprocessing__outliers__outliers__remove=False;; score=-662316.749  
total time= 0.4s
```

```
[CV 3/4; 2/16] START knn__n_neighbors=2, knn__weights=distance,  
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/  
_param_validation.py:591: FutureWarning: Passing an int for a boolean  
parameter is deprecated in version 1.2 and won't be supported anymore  
in version 1.4.
```

```
warnings.warn(
```

```
[CV 3/4; 2/16] END knn__n_neighbors=2, knn__weights=distance,  
preprocessing__outliers__outliers__remove=False;; score=-600716.239
```

```

total time= 0.4s
[CV 4/4; 2/16] START knn_n_neighbors=2, knn_weights=distance,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 2/16] END knn_n_neighbors=2, knn_weights=distance,
preprocessing_outliers_outliers_remove=False;, score=-674938.845
total time= 0.4s
[CV 1/4; 3/16] START knn_n_neighbors=3, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 1/4; 3/16] END knn_n_neighbors=3, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False;, score=-658298.596
total time= 0.4s
[CV 2/4; 3/16] START knn_n_neighbors=3, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 2/4; 3/16] END knn_n_neighbors=3, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False;, score=-747106.252
total time= 0.4s
[CV 3/4; 3/16] START knn_n_neighbors=3, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

```



```

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 3/4; 3/16] END knn__n_neighbors=3, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;, score=-616445.245
total time= 0.5s
[CV 4/4; 3/16] START knn__n_neighbors=3, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 3/16] END knn__n_neighbors=3, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;, score=-704762.013
total time= 0.4s
[CV 1/4; 4/16] START knn__n_neighbors=3, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 1/4; 4/16] END knn__n_neighbors=3, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;, score=-606764.507
total time= 0.5s
[CV 2/4; 4/16] START knn__n_neighbors=3, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

```

```

[CV 2/4; 4/16] END knn__n_neighbors=3, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-649850.891
total time= 0.5s
[CV 3/4; 4/16] START knn__n_neighbors=3, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(

[CV 3/4; 4/16] END knn__n_neighbors=3, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-553857.344
total time= 0.4s
[CV 4/4; 4/16] START knn__n_neighbors=3, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(

[CV 4/4; 4/16] END knn__n_neighbors=3, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-646927.400
total time= 0.5s
[CV 1/4; 5/16] START knn__n_neighbors=4, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(

[CV 1/4; 5/16] END knn__n_neighbors=4, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-681432.867
total time= 0.4s
[CV 2/4; 5/16] START knn__n_neighbors=4, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s

```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 2/4; 5/16] END knn__n_neighbors=4, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-754585.473
total time= 0.4s
```

```
[CV 3/4; 5/16] START knn__n_neighbors=4, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 3/4; 5/16] END knn__n_neighbors=4, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-642175.211
total time= 0.4s
```

```
[CV 4/4; 5/16] START knn__n_neighbors=4, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 4/4; 5/16] END knn__n_neighbors=4, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-724530.308
total time= 0.4s
```

```
[CV 1/4; 6/16] START knn__n_neighbors=4, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
```

```

in version 1.4.
    warnings.warn(

[CV 1/4; 6/16] END knn__n_neighbors=4, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-603579.283
total time= 0.6s
[CV 2/4; 6/16] START knn__n_neighbors=4, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 2/4; 6/16] END knn__n_neighbors=4, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-648746.782
total time= 0.4s
[CV 3/4; 6/16] START knn__n_neighbors=4, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 3/4; 6/16] END knn__n_neighbors=4, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-557267.140
total time= 0.4s
[CV 4/4; 6/16] START knn__n_neighbors=4, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 6/16] END knn__n_neighbors=4, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-644347.986
total time= 0.4s

```

```
[CV 1/4; 7/16] START knn_n_neighbors=5, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 1/4; 7/16] END knn_n_neighbors=5, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False;, score=-737094.402
total time= 0.5s
```

```
[CV 2/4; 7/16] START knn_n_neighbors=5, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 2/4; 7/16] END knn_n_neighbors=5, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False;, score=-785039.577
total time= 0.4s
```

```
[CV 3/4; 7/16] START knn_n_neighbors=5, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 3/4; 7/16] END knn_n_neighbors=5, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False;, score=-685572.592
total time= 0.4s
```

```
[CV 4/4; 7/16] START knn_n_neighbors=5, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 7/16] END knn__n_neighbors=5, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-756601.646
total time= 0.5s
[CV 1/4; 8/16] START knn__n_neighbors=5, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 1/4; 8/16] END knn__n_neighbors=5, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-625310.213
total time= 0.5s
[CV 2/4; 8/16] START knn__n_neighbors=5, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 2/4; 8/16] END knn__n_neighbors=5, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-658133.312
total time= 0.5s
[CV 3/4; 8/16] START knn__n_neighbors=5, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

```

```
[CV 3/4; 8/16] END knn__n_neighbors=5, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-571204.337
total time= 0.5s
[CV 4/4; 8/16] START knn__n_neighbors=5, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
warnings.warn(
```

```
[CV 4/4; 8/16] END knn__n_neighbors=5, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-655816.433
total time= 0.5s
[CV 1/4; 9/16] START knn__n_neighbors=6, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
warnings.warn(
```

```
[CV 1/4; 9/16] END knn__n_neighbors=6, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-780668.687
total time= 0.5s
[CV 2/4; 9/16] START knn__n_neighbors=6, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
warnings.warn(
```

```
[CV 2/4; 9/16] END knn__n_neighbors=6, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-829927.193
total time= 0.4s
[CV 3/4; 9/16] START knn__n_neighbors=6, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(
```

```
[CV 3/4; 9/16] END knn__n_neighbors=6, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;, score=-733650.379
total time= 0.6s
```

```
[CV 4/4; 9/16] START knn__n_neighbors=6, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(
```

```
[CV 4/4; 9/16] END knn__n_neighbors=6, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;, score=-802119.209
total time= 0.5s
```

```
[CV 1/4; 10/16] START knn__n_neighbors=6, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(
```

```
[CV 1/4; 10/16] END knn__n_neighbors=6, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;, score=-642579.890
total time= 0.5s
```

```
[CV 2/4; 10/16] START knn__n_neighbors=6, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
```



```

in version 1.4.
    warnings.warn(

[CV 2/4; 10/16] END knn__n_neighbors=6, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-677393.309
total time= 0.5s
[CV 3/4; 10/16] START knn__n_neighbors=6, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 3/4; 10/16] END knn__n_neighbors=6, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-590298.014
total time= 0.5s
[CV 4/4; 10/16] START knn__n_neighbors=6, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 10/16] END knn__n_neighbors=6, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-672768.941
total time= 0.5s
[CV 1/4; 11/16] START knn__n_neighbors=7, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 1/4; 11/16] END knn__n_neighbors=7, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-816080.820
total time= 0.6s

```

```
[CV 2/4; 11/16] START knn_n_neighbors=7, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 2/4; 11/16] END knn_n_neighbors=7, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False;, score=-863931.239
total time= 0.5s
```

```
[CV 3/4; 11/16] START knn_n_neighbors=7, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 3/4; 11/16] END knn_n_neighbors=7, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False;, score=-780544.808
total time= 0.6s
```

```
[CV 4/4; 11/16] START knn_n_neighbors=7, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 4/4; 11/16] END knn_n_neighbors=7, knn_weights=uniform,
preprocessing_outliers_outliers_remove=False;, score=-828585.330
total time= 0.6s
```

```
[CV 1/4; 12/16] START knn_n_neighbors=7, knn_weights=distance,
preprocessing_outliers_outliers_remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s
```

```

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 1/4; 12/16] END knn__n_neighbors=7, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;, score=-659992.985
total time= 0.6s
[CV 2/4; 12/16] START knn__n_neighbors=7, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 2/4; 12/16] END knn__n_neighbors=7, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;, score=-695564.476
total time= 0.5s
[CV 3/4; 12/16] START knn__n_neighbors=7, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 3/4; 12/16] END knn__n_neighbors=7, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;, score=-613350.520
total time= 0.5s
[CV 4/4; 12/16] START knn__n_neighbors=7, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

```

```
[CV 4/4; 12/16] END knn__n_neighbors=7, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-678432.027
total time= 0.5s
[CV 1/4; 13/16] START knn__n_neighbors=8, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
warnings.warn(
```

```
[CV 1/4; 13/16] END knn__n_neighbors=8, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-854066.959
total time= 0.6s
[CV 2/4; 13/16] START knn__n_neighbors=8, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
warnings.warn(
```

```
[CV 2/4; 13/16] END knn__n_neighbors=8, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-902788.140
total time= 0.5s
[CV 3/4; 13/16] START knn__n_neighbors=8, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
warnings.warn(
```

```
[CV 3/4; 13/16] END knn__n_neighbors=8, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-819705.387
total time= 0.6s
[CV 4/4; 13/16] START knn__n_neighbors=8, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(
```

```
[CV 4/4; 13/16] END knn__n_neighbors=8, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;, score=-873142.360
total time= 1.0s
```

```
[CV 1/4; 14/16] START knn__n_neighbors=8, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(
```

```
[CV 1/4; 14/16] END knn__n_neighbors=8, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;, score=-682468.292
total time= 0.9s
```

```
[CV 2/4; 14/16] START knn__n_neighbors=8, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(
```

```
[CV 2/4; 14/16] END knn__n_neighbors=8, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;, score=-714099.576
total time= 0.7s
```

```
[CV 3/4; 14/16] START knn__n_neighbors=8, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
```

```

in version 1.4.
    warnings.warn(

[CV 3/4; 14/16] END knn__n_neighbors=8, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-632818.248
total time= 0.9s
[CV 4/4; 14/16] START knn__n_neighbors=8, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 14/16] END knn__n_neighbors=8, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-696397.080
total time= 0.8s
[CV 1/4; 15/16] START knn__n_neighbors=9, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 1/4; 15/16] END knn__n_neighbors=9, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-874026.916
total time= 0.9s
[CV 2/4; 15/16] START knn__n_neighbors=9, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 2/4; 15/16] END knn__n_neighbors=9, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-943686.423
total time= 0.5s

```

```
[CV 3/4; 15/16] START knn__n_neighbors=9, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 3/4; 15/16] END knn__n_neighbors=9, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-850894.182
total time= 0.5s
```

```
[CV 4/4; 15/16] START knn__n_neighbors=9, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 4/4; 15/16] END knn__n_neighbors=9, knn__weights=uniform,
preprocessing__outliers__outliers__remove=False;; score=-915628.087
total time= 0.6s
```

```
[CV 1/4; 16/16] START knn__n_neighbors=9, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 1/4; 16/16] END knn__n_neighbors=9, knn__weights=distance,
preprocessing__outliers__outliers__remove=False;; score=-697455.435
total time= 0.6s
```

```
[CV 2/4; 16/16] START knn__n_neighbors=9, knn__weights=distance,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean parameter is deprecated in version 1.2 and won't be supported anymore in version 1.4.
```

```
warnings.warn(
```

```
[CV 2/4; 16/16] END knn__n_neighbors=9, knn__weights=distance, preprocessing__outliers__outliers__remove=False;; score=-732891.152 total time= 0.6s
```

```
[CV 3/4; 16/16] START knn__n_neighbors=9, knn__weights=distance, preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean parameter is deprecated in version 1.2 and won't be supported anymore in version 1.4.
```

```
warnings.warn(
```

```
[CV 3/4; 16/16] END knn__n_neighbors=9, knn__weights=distance, preprocessing__outliers__outliers__remove=False;; score=-651469.887 total time= 0.7s
```

```
[CV 4/4; 16/16] START knn__n_neighbors=9, knn__weights=distance, preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean parameter is deprecated in version 1.2 and won't be supported anymore in version 1.4.
```

```
warnings.warn(
```

```
[CV 4/4; 16/16] END knn__n_neighbors=9, knn__weights=distance, preprocessing__outliers__outliers__remove=False;; score=-716136.977 total time= 0.6s
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean parameter is deprecated in version 1.2 and won't be supported anymore in version 1.4.
```

```
warnings.warn(
```

```
GridSearchCV(cv=4,  
             estimator=Pipeline(steps=[('preprocessing',
```



```

Pipeline(steps=[('outliers',
Pipeline(steps=[('outliers',
OutlierRemover(columns=['carat',
'x',
'y',
'z',
'depth',
'table'],
remove=False)))])),
('num',
ColumnTransformer(remainder='passthrough',
transformers=[('scaler',
MinMaxScaler(),
['carat',
'x',
'y',
'z',
'depth',
'table'])]),
verbose_feature_names_out=...
OrdinalEncoder(categories=[['I1',
'SI2',
'SI1',
'VS2',
'VS1',

```

```

'VVS2',
'VVS1',
'IF']]]))],
['clarity']]],
verbose_feature_names_out=False)),
(verbose=4)),
('knn',
KNeighborsRegressor()))],
param_grid={'knn__n_neighbors': range(2, 10),
            'knn__weights': ['uniform', 'distance'],
            'preprocessing__outliers__outliers__remove':
[False]},
            scoring='neg_mean_squared_error', verbose=10)

knn = gs_knn.best_estimator_
gs_knn.best_params_
{'knn__n_neighbors': 4,
 'knn__weights': 'distance',
 'preprocessing__outliers__outliers__remove': False}

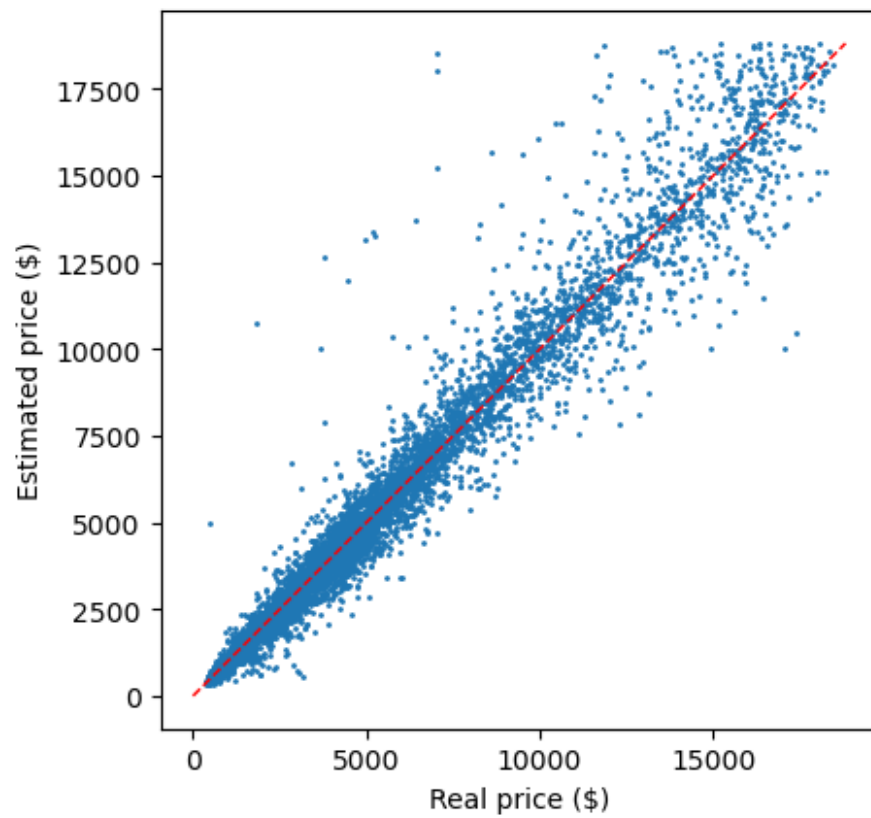
y_pred = knn.predict(X_test)

print("Mean Squared Error      : ", mean_squared_error(y_test,
y_pred))
print("Root Mean Squared Error : ",
math.sqrt(mean_squared_error(y_test, y_pred)))
print("Mean Absolute Error      : ", mean_absolute_error(y_test,
y_pred))
print("R2                        : ", r2_score(y_test, y_pred))

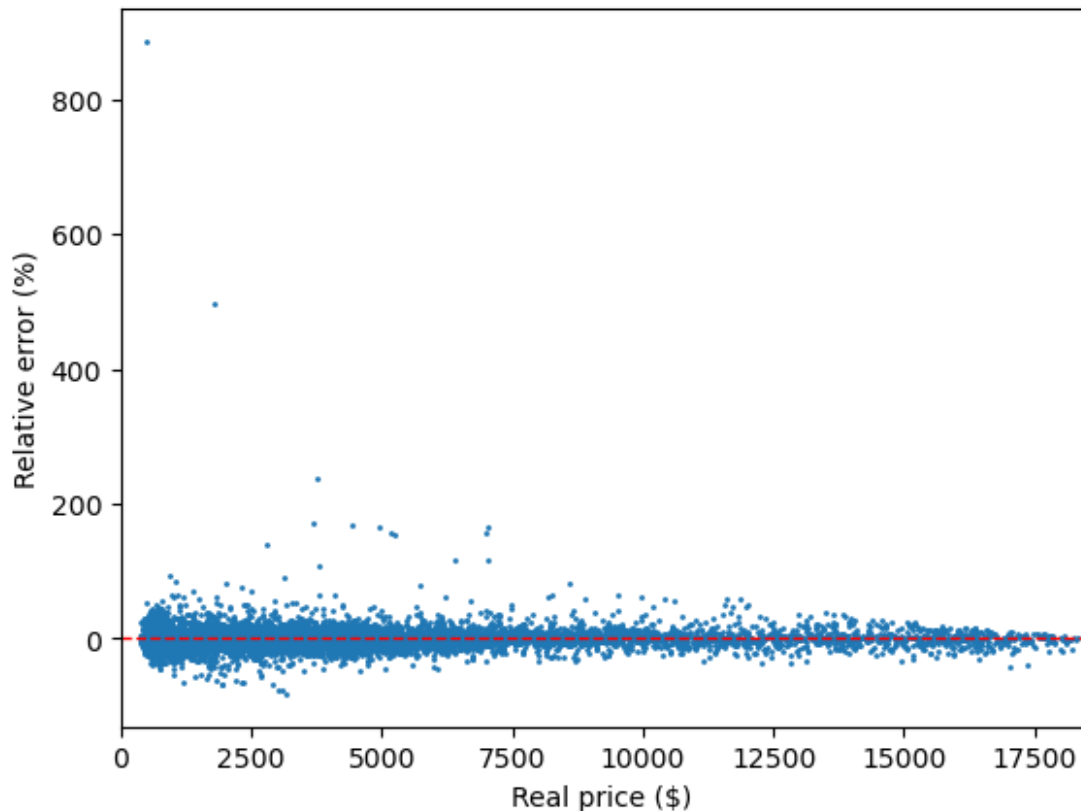
Mean Squared Error      :  525026.7948105916
Root Mean Squared Error :  724.5873272495121
Mean Absolute Error      :  352.4030351496844
R2                        :  0.9673842611908018

plot_pred_vs_test(y_test,y_pred)

```



```
plot_pred_vs_test_relative(y_test,y_pred)
```



Régression un arbre de décision

```
from sklearn.tree import DecisionTreeRegressor

pipeline_final_tree = Pipeline(steps = [
    ("preprocessing", pipeline),
    ("tree", DecisionTreeRegressor())
])

hyperparameters_tree = {
    "preprocessing__outliers__outliers_remove": [False],
    "tree__criterion" : ["squared_error", "friedman_mse",
    "absolute_error"],
    "tree__max_depth" : range(5,10),
}

gs_tree = GridSearchCV(
    estimator=pipeline_final_tree,
    param_grid=hyperparameters_tree,
    cv = 4,
    scoring = "neg_mean_squared_error",
    verbose = 10
)
```

```
gs_tree.fit(X_train, y_train)
```

```
Fitting 4 folds for each of 15 candidates, totalling 60 fits
```

```
[CV 1/4; 1/15] START preprocessing_outliers__outliers__remove=False,  
tree__criterion=squared_error, tree__max_depth=5
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
[CV 1/4; 1/15] END preprocessing_outliers__outliers__remove=False,  
tree__criterion=squared_error, tree__max_depth=5;; score=-965219.829  
total time= 0.1s
```

```
[CV 2/4; 1/15] START preprocessing_outliers__outliers__remove=False,  
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[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
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[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
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```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/  
_param_validation.py:591: FutureWarning: Passing an int for a boolean  
parameter is deprecated in version 1.2 and won't be supported anymore  
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warnings.warn(  
[CV 2/4; 1/15] END preprocessing_outliers__outliers__remove=False,  
tree__criterion=squared_error, tree__max_depth=5;; score=-976181.804  
total time= 0.1s
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[CV 3/4; 1/15] START preprocessing_outliers__outliers__remove=False,  
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[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
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[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
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```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
[CV 3/4; 1/15] END preprocessing_outliers__outliers__remove=False,  
tree__criterion=squared_error, tree__max_depth=5;; score=-967575.901  
total time= 0.1s
```

```
[CV 4/4; 1/15] START preprocessing_outliers__outliers__remove=False,  
tree__criterion=squared_error, tree__max_depth=5
```

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[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
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[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
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total time= 0.1s
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```

```
warnings.warn(
```

```
[CV 4/4; 1/15] END preprocessing_outliers_outliers_remove=False, tree_criterion=squared_error, tree_max_depth=5;, score=-1089352.461 total time= 0.1s
```

```
[CV 1/4; 2/15] START preprocessing_outliers_outliers_remove=False, tree_criterion=squared_error, tree_max_depth=6
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
[CV 1/4; 2/15] END preprocessing_outliers_outliers_remove=False, tree_criterion=squared_error, tree_max_depth=6;, score=-716597.813 total time= 0.1s
```

```
[CV 2/4; 2/15] START preprocessing_outliers_outliers_remove=False, tree_criterion=squared_error, tree_max_depth=6
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
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/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean parameter is deprecated in version 1.2 and won't be supported anymore in version 1.4.
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```

```
warnings.warn(
```

```
[CV 2/4; 2/15] END preprocessing_outliers_outliers_remove=False, tree_criterion=squared_error, tree_max_depth=6;, score=-752050.608 total time= 0.1s
```

```
[CV 3/4; 2/15] START preprocessing_outliers_outliers_remove=False, tree_criterion=squared_error, tree_max_depth=6
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
[CV 3/4; 2/15] END preprocessing_outliers_outliers_remove=False, tree_criterion=squared_error, tree_max_depth=6;, score=-731692.880 total time= 0.1s
```

```
[CV 4/4; 2/15] START preprocessing_outliers_outliers_remove=False, tree_criterion=squared_error, tree_max_depth=6
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
```

```
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
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[CV 4/4; 2/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=6;; score=-822013.415
total time= 0.1s
[CV 1/4; 3/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=7
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 1/4; 3/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=7;; score=-572110.565
total time= 0.1s
[CV 2/4; 3/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=7
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
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  warnings.warn(

[CV 2/4; 3/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=7;; score=-612096.788
total time= 0.1s
[CV 3/4; 3/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=7
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 3/4; 3/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=7;; score=-582646.157
total time= 0.2s

```

```

[CV 4/4; 3/15] START preprocessing_outliers_outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=7
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean
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[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 4/4; 3/15] END preprocessing_outliers_outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=7;; score=-647112.343
total time= 0.2s
[CV 1/4; 4/15] START preprocessing_outliers_outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=8
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

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  warnings.warn(

[CV 1/4; 4/15] END preprocessing_outliers_outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=8;; score=-508989.434
total time= 0.2s
[CV 2/4; 4/15] START preprocessing_outliers_outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=8
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 2/4; 4/15] END preprocessing_outliers_outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=8;; score=-495164.067
total time= 0.1s
[CV 3/4; 4/15] START preprocessing_outliers_outliers__remove=False,
tree__criterion=squared_error, tree__max_depth=8

```



```

[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

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[CV 3/4; 4/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=squared_error, tree_max_depth=8;, score=-490653.799
total time= 0.1s
[CV 4/4; 4/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=squared_error, tree_max_depth=8
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 4/4; 4/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=squared_error, tree_max_depth=8;, score=-540577.112
total time= 0.1s
[CV 1/4; 5/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=squared_error, tree_max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

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[CV 1/4; 5/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=squared_error, tree_max_depth=9;, score=-447074.185
total time= 0.2s
[CV 2/4; 5/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=squared_error, tree_max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s

```

```

[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 2/4; 5/15] END preprocessing_outliers_outliers_remove=False,
tree__criterion=squared_error, tree__max_depth=9;, score=-448487.591
total time= 0.2s
[CV 3/4; 5/15] START preprocessing_outliers_outliers_remove=False,
tree__criterion=squared_error, tree__max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
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[CV 3/4; 5/15] END preprocessing_outliers_outliers_remove=False,
tree__criterion=squared_error, tree__max_depth=9;, score=-414040.914
total time= 0.2s
[CV 4/4; 5/15] START preprocessing_outliers_outliers_remove=False,
tree__criterion=squared_error, tree__max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 4/4; 5/15] END preprocessing_outliers_outliers_remove=False,
tree__criterion=squared_error, tree__max_depth=9;, score=-493849.286
total time= 0.2s
[CV 1/4; 6/15] START preprocessing_outliers_outliers_remove=False,
tree__criterion=friedman_mse, tree__max_depth=5
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
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[CV 1/4; 6/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=friedman_mse, tree_max_depth=5;, score=-965219.829
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[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 4/4; 6/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=friedman_mse, tree_max_depth=5;, score=-1089840.097
total time= 0.1s
[CV 1/4; 7/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=friedman_mse, tree_max_depth=6
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
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```

    warnings.warn(

```

```

[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 1/4; 7/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=6;; score=-716597.813
total time= 0.2s

```

```

[CV 2/4; 7/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=6

```

```

[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s

```

```

[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s

```

```

[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

```

```

[CV 2/4; 7/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=6;; score=-752045.130
total time= 0.1s

```

```

[CV 3/4; 7/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=6

```

```

[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s

```

```

[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s

```

```

[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

```

```

[CV 3/4; 7/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=6;; score=-731692.880
total time= 0.1s

```

```

[CV 4/4; 7/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=6

```

```

[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s

```

```

[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s

```

```

[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

```

```

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/

```

```

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1.4.

```

```

    warnings.warn(

```

```

[CV 4/4; 7/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=6;; score=-821579.622
total time= 0.1s

```

```

[CV 1/4; 8/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=7

```

```

[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s

```

```

[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s

```

```

[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

```

```

[CV 1/4; 8/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=7;; score=-572110.565
total time= 0.1s
[CV 2/4; 8/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=7
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

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dation.py:591: FutureWarning: Passing an int for a boolean parameter
is deprecated in version 1.2 and won't be supported anymore in version
1.4.
    warnings.warn(

[CV 2/4; 8/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=7;; score=-609635.493
total time= 0.1s
[CV 3/4; 8/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=7
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 3/4; 8/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=7;; score=-583002.593
total time= 0.1s
[CV 4/4; 8/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=7
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_vali
dation.py:591: FutureWarning: Passing an int for a boolean parameter
is deprecated in version 1.2 and won't be supported anymore in version
1.4.
    warnings.warn(

[CV 4/4; 8/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=7;; score=-647112.343

```

```

total time= 0.1s
[CV 1/4; 9/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=8
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 1/4; 9/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=8;; score=-508987.393
total time= 0.2s
[CV 2/4; 9/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=8
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_vali
dation.py:591: FutureWarning: Passing an int for a boolean parameter
is deprecated in version 1.2 and won't be supported anymore in version
1.4.
    warnings.warn(

[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 2/4; 9/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=8;; score=-499739.474
total time= 0.2s
[CV 3/4; 9/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=8
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_vali
dation.py:591: FutureWarning: Passing an int for a boolean parameter
is deprecated in version 1.2 and won't be supported anymore in version
1.4.
    warnings.warn(

[CV 3/4; 9/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=8;; score=-489598.582
total time= 0.1s
[CV 4/4; 9/15] START preprocessing__outliers__outliers__remove=False,

```

```

tree__criterion=friedman_mse, tree__max_depth=8
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 4/4; 9/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=8;;, score=-539527.673
total time= 0.1s
[CV 1/4; 10/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_vali
dation.py:591: FutureWarning: Passing an int for a boolean parameter
is deprecated in version 1.2 and won't be supported anymore in version
1.4.
  warnings.warn(

[CV 1/4; 10/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=9;;, score=-446758.605
total time= 0.1s
[CV 2/4; 10/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 2/4; 10/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=9;;, score=-456404.546
total time= 0.2s
[CV 3/4; 10/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/_param_vali
dation.py:591: FutureWarning: Passing an int for a boolean parameter
is deprecated in version 1.2 and won't be supported anymore in version

```

```

1.4.
    warnings.warn(

[CV 3/4; 10/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=9;, score=-420013.041
total time= 0.1s
[CV 4/4; 10/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 4/4; 10/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=friedman_mse, tree__max_depth=9;, score=-496000.067
total time= 0.2s
[CV 1/4; 11/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=5
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
[CV 1/4; 11/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=5;, score=-1077178.216
total time= 26.3s
[CV 2/4; 11/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=5
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 2/4; 11/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=5;, score=-1128214.717
total time= 31.2s
[CV 3/4; 11/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=5
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

```



```

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 3/4; 11/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=5;, score=-1082601.228
total time= 33.8s
[CV 4/4; 11/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=5
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 11/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=5;, score=-1218747.420
total time= 35.4s
[CV 1/4; 12/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=6
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 1/4; 12/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=6;, score=-815282.602
total time= 41.6s
[CV 2/4; 12/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=6
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

```

```

[CV 2/4; 12/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=6;, score=-872193.272
total time= 37.2s
[CV 3/4; 12/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=6
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(

[CV 3/4; 12/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=6;, score=-851702.806
total time= 37.4s
[CV 4/4; 12/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=6
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(

[CV 4/4; 12/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=6;, score=-950914.989
total time= 39.2s
[CV 1/4; 13/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=7
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(

[CV 1/4; 13/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=7;, score=-592958.821
total time= 37.5s
[CV 2/4; 13/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=7
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s

```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(
```

```
[CV 2/4; 13/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=7;, score=-687787.930
total time= 35.6s
```

```
[CV 3/4; 13/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=7
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(
```

```
[CV 3/4; 13/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=7;, score=-644889.079
total time= 35.0s
```

```
[CV 4/4; 13/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=7
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
  warnings.warn(
```

```
[CV 4/4; 13/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=7;, score=-691344.530
total time= 35.1s
```

```
[CV 1/4; 14/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=8
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
```

```

in version 1.4.
    warnings.warn(

[CV 1/4; 14/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=8;, score=-503145.652
total time= 37.8s
[CV 2/4; 14/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=8
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 2/4; 14/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=8;, score=-569628.230
total time= 40.8s
[CV 3/4; 14/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=8
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 3/4; 14/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=8;, score=-540345.072
total time= 36.7s
[CV 4/4; 14/15] START preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=8
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 14/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=8;, score=-604075.077
total time= 34.2s

```

```
[CV 1/4; 15/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 1/4; 15/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=9;, score=-457293.364
total time= 35.6s
```

```
[CV 2/4; 15/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 2/4; 15/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=9;, score=-511685.337
total time= 45.2s
```

```
[CV 3/4; 15/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
```

```
warnings.warn(
```

```
[CV 3/4; 15/15] END preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=9;, score=-481074.315
total time= 41.3s
```

```
[CV 4/4; 15/15] START preprocessing_outliers_outliers_remove=False,
tree_criterion=absolute_error, tree_max_depth=9
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s
```

```

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 15/15] END preprocessing__outliers__outliers__remove=False,
tree__criterion=absolute_error, tree__max_depth=9;; score=-535514.070
total time= 40.6s
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

GridSearchCV(cv=4,
              estimator=Pipeline(steps=[('preprocessing',
                                         Pipeline(steps=[('outliers',
                                                             Pipeline(steps=[('outliers',
OutlierRemover(columns=['carat',
                        'x',
                        'y',
                        'z',
                        'depth',
                        'table'],
                        remove=False)))])),
                                         ('num',
ColumnTransformer(remainder='passthrough',
transformers=[('scaler',
MinMaxScaler(),
               ['carat',
                'x',

```

```

'y',
'z',
'depth',
'table']]],
verbose_feature_names_out=...
'SI2',
'SI1',
'VS2',
'VS1',
'VVS2',
'VVS1',
'IF']]]))],
['clarity']]],
verbose_feature_names_out=False))],
(verbose=4)),
('tree',
DecisionTreeRegressor()))],
param_grid={'preprocessing__outliers__outliers__remove':
[False],
'tree__criterion': ['squared_error',
'friedman_mse',
'absolute_error'],
'tree__max_depth': range(5, 10)},
scoring='neg_mean_squared_error', verbose=10)

tree = gs_tree.best_estimator_
gs_tree.best_params_
{'preprocessing__outliers__outliers__remove': False,
'tree__criterion': 'squared_error',
'tree__max_depth': 9}

y_pred = tree.predict(X_test)

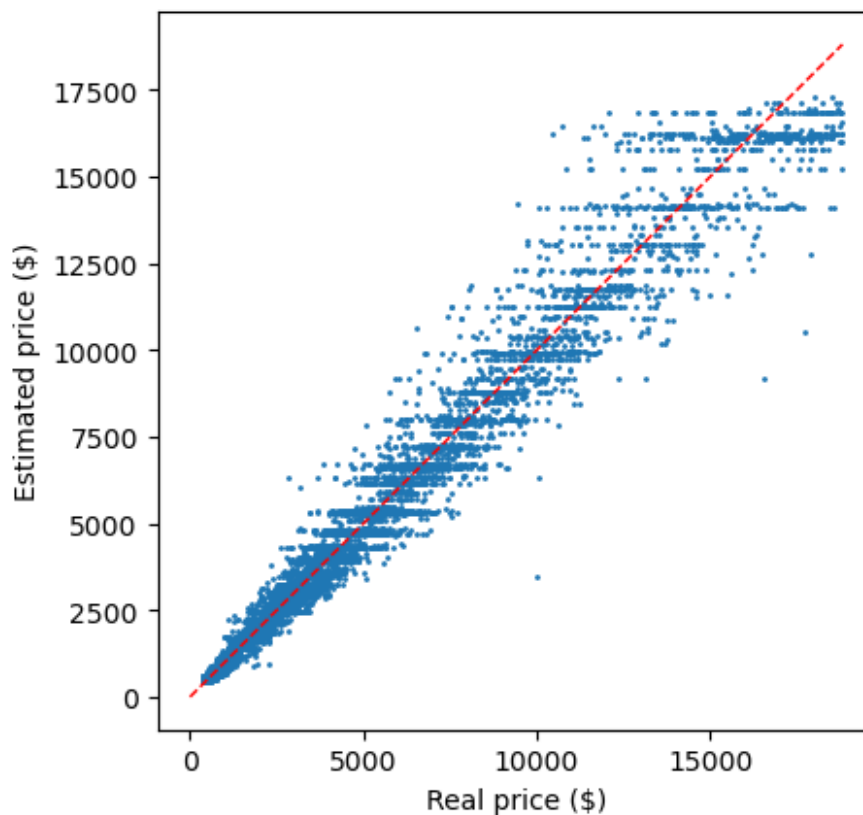
print("Mean Squared Error : ", mean_squared_error(y_test,
y_pred))

```

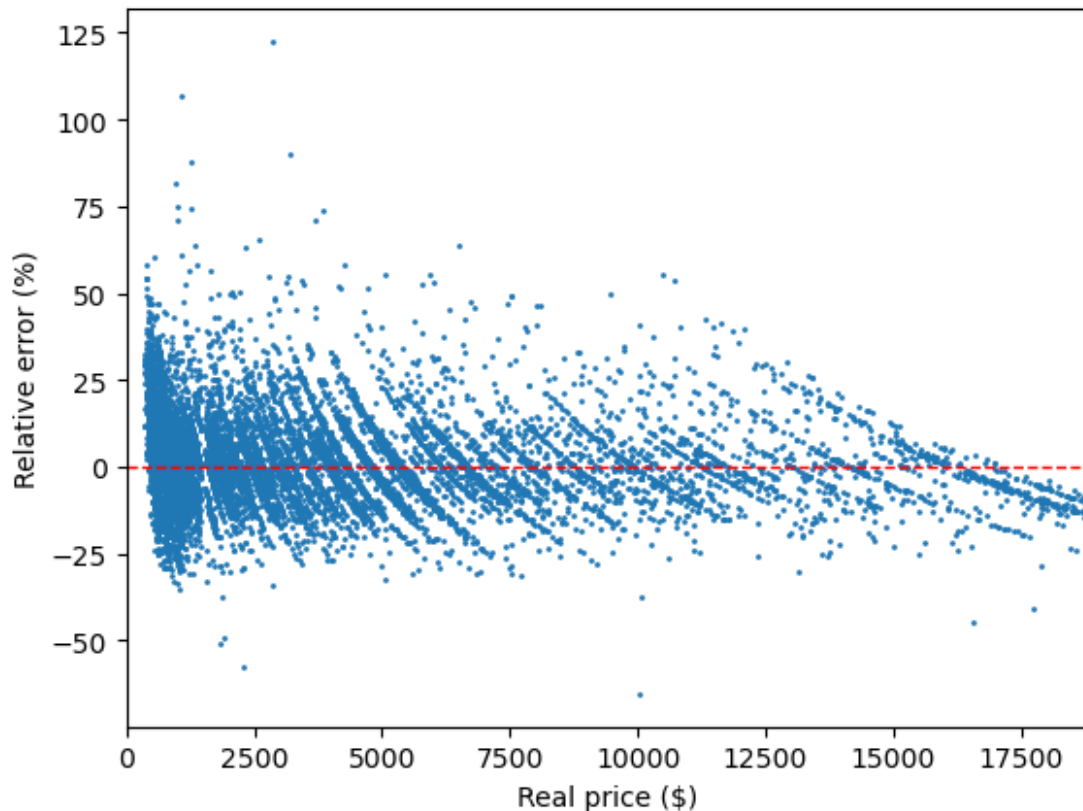
```
print("Root Mean Squared Error : ",  
      math.sqrt(mean_squared_error(y_test, y_pred)))  
print("Mean Absolute Error      : ", mean_absolute_error(y_test,  
y_pred))  
print("R2                        : ", r2_score(y_test, y_pred))
```

```
Mean Squared Error      : 429901.2981309896  
Root Mean Squared Error : 655.6685886413879  
Mean Absolute Error     : 355.02808473903065  
R2                      : 0.9732936516913694
```

```
plot_pred_vs_test(y_pred, y_test)
```



```
plot_pred_vs_test_relative(y_pred, y_test)
```

Régression avec une forêt aléatoire

```
from sklearn.ensemble import RandomForestRegressor

pipeline_final_forest = Pipeline(steps = [
    ("preprocessing", pipeline),
    ("forest", RandomForestRegressor())
])

hyperparameters_forest = {
    "preprocessing_outliers_outliers_remove": [False],
    "forest_n_estimators" : [50, 100],
    "forest_max_depth" : range(5,7),
}

gs_forest = GridSearchCV(
    estimator=pipeline_final_forest,
    param_grid=hyperparameters_forest,
    cv = 4,
    scoring = "neg_mean_squared_error",
    verbose = 10
)

gs_forest.fit(X_train, y_train)
```

```

Fitting 4 folds for each of 4 candidates, totalling 16 fits
[CV 1/4; 1/4] START forest__max_depth=5, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 1/4; 1/4] END forest__max_depth=5, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False;; score=-804831.925
total time= 1.9s
[CV 2/4; 1/4] START forest__max_depth=5, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 2/4; 1/4] END forest__max_depth=5, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False;; score=-817061.467
total time= 1.8s
[CV 3/4; 1/4] START forest__max_depth=5, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 3/4; 1/4] END forest__max_depth=5, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False;; score=-802302.195
total time= 1.8s
[CV 4/4; 1/4] START forest__max_depth=5, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

```

```

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 1/4] END forest__max_depth=5, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False;; score=-932413.735
total time= 1.8s
[CV 1/4; 2/4] START forest__max_depth=5, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 1/4; 2/4] END forest__max_depth=5, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False;; score=-792904.397
total time= 3.8s
[CV 2/4; 2/4] START forest__max_depth=5, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 2/4; 2/4] END forest__max_depth=5, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False;; score=-827616.664
total time= 4.3s
[CV 3/4; 2/4] START forest__max_depth=5, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

```

```
[CV 3/4; 2/4] END forest__max_depth=5, forest__n_estimators=100,  
preprocessing__outliers__outliers_remove=False;; score=-808416.534  
total time= 4.3s  
[CV 4/4; 2/4] START forest__max_depth=5, forest__n_estimators=100,  
preprocessing__outliers__outliers_remove=False  
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s  
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s  
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/  
_param_validation.py:591: FutureWarning: Passing an int for a boolean  
parameter is deprecated in version 1.2 and won't be supported anymore  
in version 1.4.  
warnings.warn(
```

```
[CV 4/4; 2/4] END forest__max_depth=5, forest__n_estimators=100,  
preprocessing__outliers__outliers_remove=False;; score=-926416.057  
total time= 4.9s  
[CV 1/4; 3/4] START forest__max_depth=6, forest__n_estimators=50,  
preprocessing__outliers__outliers_remove=False  
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s  
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s  
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/  
_param_validation.py:591: FutureWarning: Passing an int for a boolean  
parameter is deprecated in version 1.2 and won't be supported anymore  
in version 1.4.  
warnings.warn(
```

```
[CV 1/4; 3/4] END forest__max_depth=6, forest__n_estimators=50,  
preprocessing__outliers__outliers_remove=False;; score=-585753.043  
total time= 2.9s  
[CV 2/4; 3/4] START forest__max_depth=6, forest__n_estimators=50,  
preprocessing__outliers__outliers_remove=False  
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s  
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s  
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/  
_param_validation.py:591: FutureWarning: Passing an int for a boolean  
parameter is deprecated in version 1.2 and won't be supported anymore  
in version 1.4.  
warnings.warn(
```

```
[CV 2/4; 3/4] END forest__max_depth=6, forest__n_estimators=50,  
preprocessing__outliers__outliers_remove=False;; score=-633002.719  
total time= 3.0s  
[CV 3/4; 3/4] START forest__max_depth=6, forest__n_estimators=50,  
preprocessing__outliers__outliers_remove=False  
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
```

```
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(
```

```
[CV 3/4; 3/4] END forest__max_depth=6, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False;; score=-611225.003
total time= 2.6s
```

```
[CV 4/4; 3/4] START forest__max_depth=6, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(
```

```
[CV 4/4; 3/4] END forest__max_depth=6, forest__n_estimators=50,
preprocessing__outliers__outliers__remove=False;; score=-692630.843
total time= 2.9s
```

```
[CV 1/4; 4/4] START forest__max_depth=6, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(
```

```
[CV 1/4; 4/4] END forest__max_depth=6, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False;; score=-580946.308
total time= 5.8s
```

```
[CV 2/4; 4/4] START forest__max_depth=6, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False
```

```
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s
```

```
/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
```

```

in version 1.4.
    warnings.warn(

[CV 2/4; 4/4] END forest__max_depth=6, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False;; score=-631259.970
total time= 5.8s
[CV 3/4; 4/4] START forest__max_depth=6, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 3/4; 4/4] END forest__max_depth=6, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False;; score=-605060.113
total time= 6.7s
[CV 4/4; 4/4] START forest__max_depth=6, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.1s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

[CV 4/4; 4/4] END forest__max_depth=6, forest__n_estimators=100,
preprocessing__outliers__outliers__remove=False;; score=-688905.987
total time= 6.3s
[Pipeline] ..... (step 1 of 3) Processing outliers, total= 0.0s
[Pipeline] ..... (step 2 of 3) Processing num, total= 0.0s
[Pipeline] ..... (step 3 of 3) Processing ord, total= 0.0s

/home/kf/.local/lib/python3.10/site-packages/sklearn/utils/
_param_validation.py:591: FutureWarning: Passing an int for a boolean
parameter is deprecated in version 1.2 and won't be supported anymore
in version 1.4.
    warnings.warn(

GridSearchCV(cv=4,
             estimator=Pipeline(steps=[('preprocessing',
                                       Pipeline(steps=[('outliers',
Pipeline(steps=[('outliers',

```

```
OutlierRemover(columns=['carat',
                        'x',
                        'y',
                        'z',
                        'depth',
                        'table'],
remove=False)))])),
                        ('num',

ColumnTransformer(remainder='passthrough',
transformers=[('scaler',
MinMaxScaler(),
['carat',
'x',
'y',
'z',
'depth',
'table'])]),
verbose_feature_names_out=...
OrdinalEncoder(categories=[['I1',
'SI2',
'SI1',
'VS2',
'VS1',
'VVS2',
'VVS1',
```

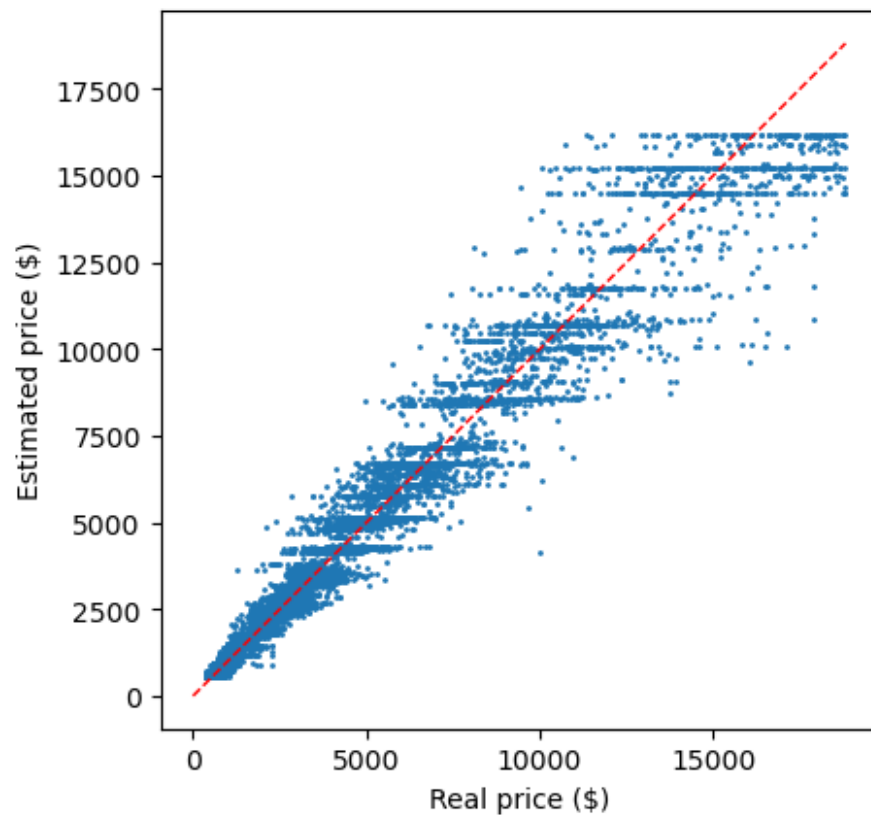
```

'IF']]]))],
['clarity']]],
verbose_feature_names_out=False))),
(verbose=4)),
('forest',
RandomForestRegressor()))],
param_grid={'forest__max_depth': range(5, 7),
            'forest__n_estimators': [50, 100],
            'preprocessing__outliers__outliers__remove':
[False]},
            scoring='neg_mean_squared_error', verbose=10)
forest = gs_forest.best_estimator_
gs_forest.best_params_
{'forest__max_depth': 6,
 'forest__n_estimators': 100,
 'preprocessing__outliers__outliers__remove': False}
y_pred = forest.predict(X_test)
print("Mean Squared Error      : ", mean_squared_error(y_test,
y_pred))
print("Root Mean Squared Error : ",
math.sqrt(mean_squared_error(y_test, y_pred)))
print("Mean Absolute Error      : ", mean_absolute_error(y_test,
y_pred))
print("R2                        : ", r2_score(y_test, y_pred))

Mean Squared Error      : 632162.947161958
Root Mean Squared Error : 795.0867544878093
Mean Absolute Error      : 446.9615962171163
R2                        : 0.9607287441835693

plot_pred_vs_test(y_pred, y_test)

```

```
plot_pred_vs_test_relative(y_pred, y_test)
```

